Table 1: The ratio of positive rewards received at different epochs on the ScanNet training set.

Epochs	50	100	200	300
Ratio of Positive Rewards (%)	60.2	65.1	69.5	70.3

Table 2: Class-agnostic object segmentation results of an additional Undo agent on the ScanNet validation set.

Method	AP(%)	AP50(%)	AP25(%)
Ours	24.2	46.2	74.7
Ours-Undo	23.9	45.0	71.4

Table 3: Class-agnostic object segmentation results of SAM3D on the ScanNet validation set.

Method	AP(%)	AP50(%)	AP25(%)
SAM3D	13.2	30.8	59.1
Ours	24.2	46.2	74.7

Table 4: Segmentation performance on the ScanNet validation set.

		AP(%)	AP50(%)	AP25(%)
Different Foundation Models	Replace DINOv2 with Concerto	22.7	45.2	75.1
	Replace TRELLIS with Hunyuan3D 2.0	24.3	46.1	75.6
	Replace TRELLIS with Direct3D	22.5	44.9	76.0
	FoundObj (Ours)	24.2	46.2	74.7
Different Rewards	FoundObj with +1/-1 rewards	23.3	45.1	75.1
	FoundObj with +5/-1 rewards	24.3	46.1	74.6
	FoundObj with +10/-1 rewards	24.2	46.2	74.7
Sparse Views	25% views	19.6	43.0	72.8
	50% views	21.6	43.9	73.3
	100% views	24.2	46.2	74.7
Camera Pose Perturbation	0% perturbation	24.2	46.2	74.7
	25% perturbation	20.6	43.1	75.5
	50% perturbation	18.7	39.0	70.3

Table 5: Open-vocabulary instance segmentation on the ScanNet validation set.

	Methods	AP(%)	AP50(%)	AP25(%)
Fully Supervised	OpenScene+Mask3D	11.7	15.2	17.8
	OpenIns3D	23.7	29.4	32.8
	OpenMask3D	15.4	19.9	23.1
Unsupervised	Ours	6.7	12.7	16.4

Table 6: Comparisons with more baselines on the ScanNet validation set.

Method	AP(%)	AP50(%)	AP25(%)
Concerto with Ncut	18.2	38.4	71.6
Concerto+DINOv2 with Ncut	19.8	41.2	72.2
TRELLIS+DINOv2 with Ncut	16.4	36.8	66.7
Ours	24.2	46.2	74.7

Table 7: The fairest and comprehensive comparisons on the ScanNet validation set.

	Methods	AP(%)	AP50(%)	AP25(%)
DINO only	Unscene3D (DINOv1)	17.7	35.6	65.2
	Part2Object (DINOv2)	**19.6**	38.4	64.9
	Ours (DINOv2)	19.5	**40.2**	**72.7**
TRELLIS only	Unscene3D	10.1	24.5	56.3
	Part2Object	12.6	29.5	65.1
	Ours	**15.3**	**37.2**	**67.6**
DINO+TRELLIS	Unscene3D (DINOv1+TRELLIS)	15.3	33.2	68.5
	Unscene3D (DINOv2+TRELLIS)	16.4	36.8	66.7
	Part2Object (DINOv2+TRELLIS)	17.7	37.5	70.9
	Ours (DINOv2+TRELLIS)	**24.2**	**46.2**	**74.7**

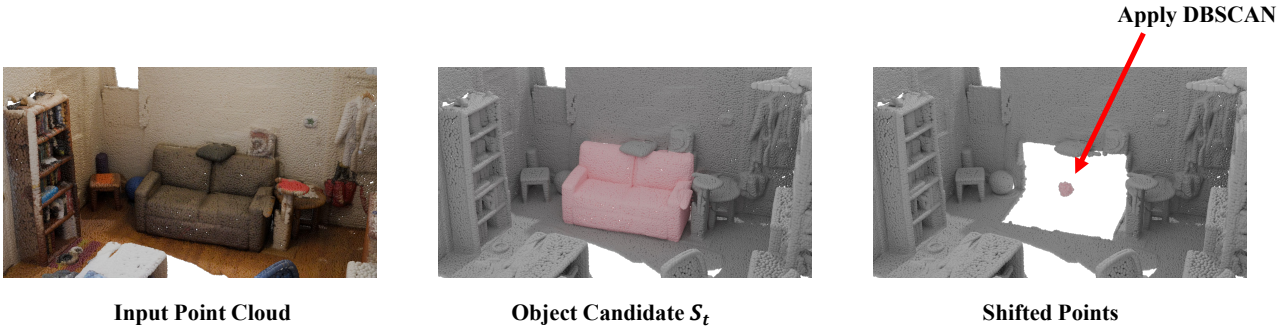

Figure 1: Visual illustration of DBSCAN.